Did the Role Fire?

Auditing the Manipulation Checks Behind Role-Prompted Multi-Agent Systems

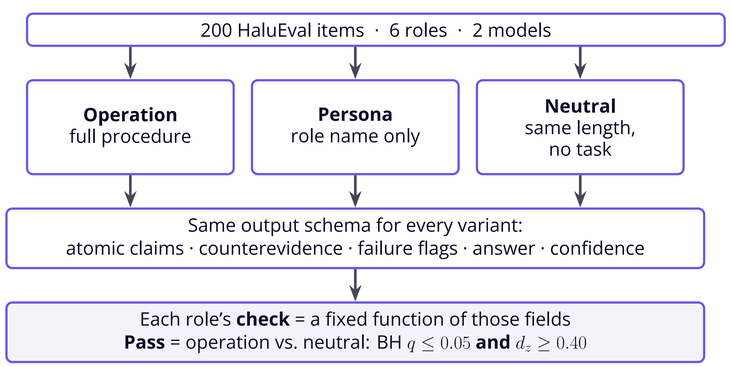
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Agent Evaluation Science
Fall 2026 · NYC
Session 1 · Poster 23

1 The question

Multi-agent systems give agents jobs like *critic*, *verifier*, or *explorer*. When the benchmark score moves, papers credit the roles. But a score can move for other reasons: prompt length, tone, abstention. **Did the role actually do its job?** Experimental psychology faces the same problem and answers it with a **manipulation check**: confirm the treatment was delivered before interpreting its effect. We build one for each of six role prompts, then audit the checks themselves.

2 Method

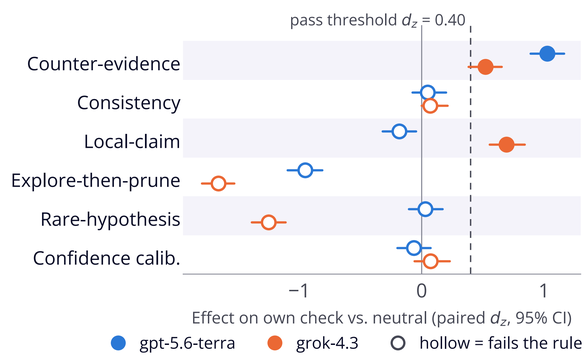


- **Task:** HaluEval QA. Given a knowledge passage, does a candidate answer contain hallucinated content? (yes/no)
- **Models:** `gpt-5.6-terra` (primary), `grok-4.3` (replication). Temperature 0, seed 1337, one sample per item.
- **Scale:** 3,600 calls per model, 0 refusals, 0 exclusions.
- **Statistics:** d_z = mean paired difference / SD of differences; 2,000-resample paired bootstrap; BH across the six roles. Rule fixed before data.

3 The six checks, and what their guards did

Role	What the check computes → what happened to its guard
Counter-evidence	# counterclaims + # claims marked unsupported or contradicted. No guard: any counterclaim counts, whatever its quality.
Consistency	Rate of contradiction or calculation-error flags − abstention rate. Defeatable: flag everything and still answer to score 1.0.
Local-claim	# claims × (1 + share with a source). Models self-report a source on 99% of claims, so the factor is 2 on 1,191 of 1,200 records: the check is 2 × claim count.
Explore-then-prune	# distinct claims × (1 − share marked uncertain). Only 0.5–1.7% of claims are uncertain, so the factor is ≈1.
Rare-hypothesis	# supported claims with a source − # without. Penalty almost never fires (1 of 7,638 claims on terra).
Confidence calibration	−ECE over stated confidences (10 equal-mass bins). Defeatable on terra: lowering every confidence by 0.05 cuts ECE from 0.144 to 0.112 (no shift helps on grok).

4 Verdicts: one role passes on terra, two on grok



terra: only counter-evidence passes ($d_z = +1.03$; counterclaims plus flags rise from 0.61 to 1.285 per answer).

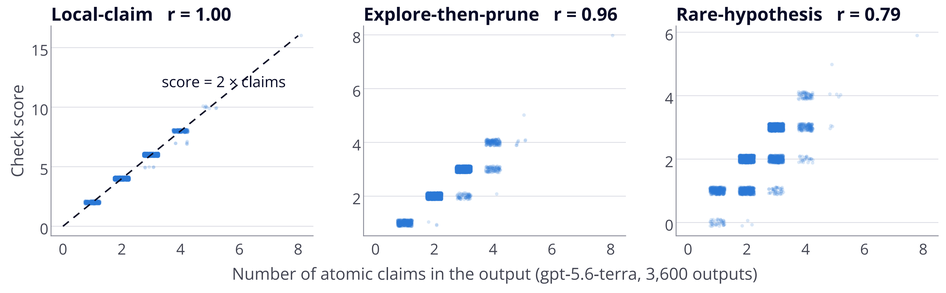
grok: counter-evidence ($+0.52$) and local-claim ($+0.69$).

Bare role names barely move their own checks: $|d_z| \le 0.30$ on both models.

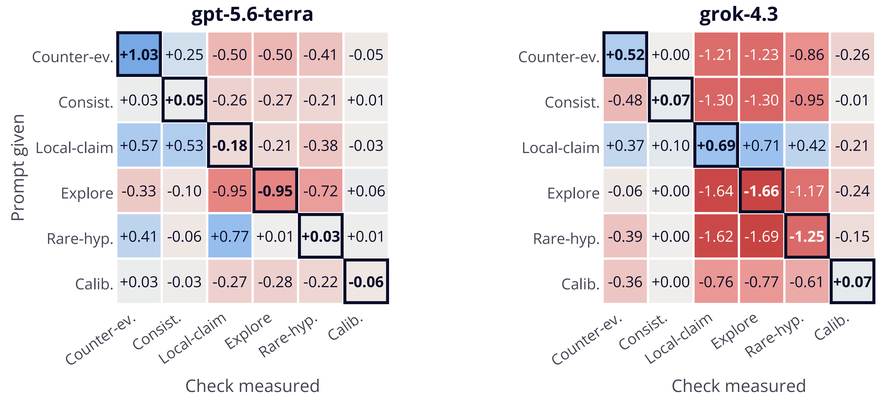
Explore-then-prune goes *down*: -0.95 and -1.66.

5 Auditing the instrument

Three checks are claim counters. Local-claim is exactly 2 × claim count on this data. Explore-then-prune and rare-hypothesis follow claim count closely. That redundancy comes from how the checks are defined, not from what the prompts do.



Do prompts move only their own check? Each row is a prompt; each column is a check (d_z vs. neutral; diagonal outlined). A role is *specific* if its own effect beats every other by ≥ 0.20, a criterion fixed before we computed the matrix. **This holds in 1 of 12 cells.**



6 What the audit means

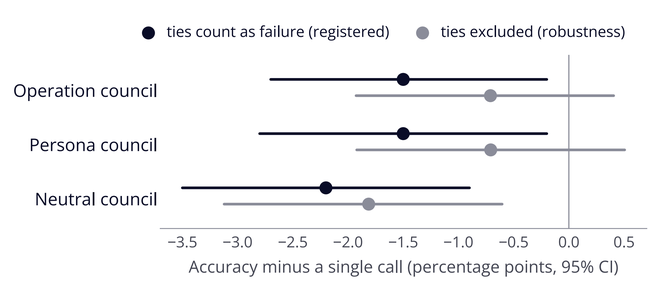
The explore-then-prune reversal is an artifact. The prompt says: *"Record the **surviving** hypotheses as atomic claims."* The check counts recorded claims as first-pass breadth, so fewer claims is what the prompt asks for, not evidence of less exploration.

Three guards were never tested, not proven broken. The behaviour they guard against (unsourced, uncertain, or fabricated claims) barely occurred. But "sourced" is the model's own report, so these guards check self-reports.

Direction matters. Counting only effects in the *same* direction (an opposite effect may be a trade-off), counter-evidence clears the margin on both models ($+0.77$, $+0.52$); nothing else does. *Sensitivity analysis added after review.*

Councils do not beat a single call

1,000 HaluEval items, `gpt-5.6-terra`, majority vote. Accuracy 0.846 / 0.846 / 0.839 vs. 0.861. Gains above 0.6 points are ruled out under both tie rules.



No role has a validated pass.
The only terra pass is an unguarded raw count; the grok local-claim pass is a claim counter. A benchmark delta does not show a role did its job, and **neither does a manipulation check until the check itself is tested.**

Next

- **Sensitivity:** preregistered positive (worked example) and negative ("avoid it") controls for every check.
- **Convergent validity:** judge or human ratings of whether counterclaims are substantive.
- **Better checks:** normalize by claim count; record first-pass hypotheses.
- **Scope:** reworded prompts and a third, open-weight model. Today: six roles, one council design, one benchmark, two models.

References. Hauser, Ellsworth & Gonzalez (2018), *Are manipulation checks necessary?* Front. Psychol. • Ejelöv & Luke (2020), *"Rarely safe to assume"*, J. Exp. Soc. Psychol. • Cemri et al. (2025), *Why do multi-agent LLM systems fail?* NeurIPS • Zheng et al. (2024), *When "A Helpful Assistant" is not really helpful*, Findings of EMNLP • Dietrich (2026), *When roles fail*, arXiv:2604.27228 • Li et al. (2023), *HaluEval*, EMNLP.

Contact: zhouk.2009@gmail.com, sparshr@mit.edu, maximkud21+research@gmail.com · Code, records and this poster: **github.com/kevzho/role-prompted-systems** · Exploratory analyses (Sections 5–6) were computed on already-observed records. · Model inference funded by Microsoft for Startups Azure credits.